

Explainable AI Financial Platform Using Attention Based Deep Learning Models

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Abstract: Financial institutions face increasing regulatory scrutiny around the deployment of opaque deep learning models for stock market prediction. This paper presents a complete Explainable AI (XAI) financial platform that pairs a stacked Long Short-Term Memory (LSTM) network with Multi-Head Attention for price forecasting, SHAP (SHapley Additive exPlanations) for feature-level attribution, and unsupervised KMeans clustering for dynamic market regime detection. The platform delivers dual-layer explainability, capturing both which technical indicators and which historical time steps influence each individual prediction. Evaluated on a held-out test set, the system achieves 68–75% directional accuracy on high-confidence signals. All predictions are fully audited in accordance with SEC and FINRA record-keeping standards. The system is deployed as a production-grade full-stack web application supporting real-time predictions for 50 large-cap stocks through a React.js dashboard backed by a FastAPI REST API.

Keywords: *Explainable AI (XAI), Long Short-Term Memory (LSTM), Multi-Head Attention, SHAP, Stock Price Forecasting, Market Regime Detection, Financial Deep Learning, KMeans Clustering.*

I. INTRODUCTION

Modern financial markets generate vast volumes of time-series data shaped by macroeconomic events, earnings releases, investor sentiment, and geopolitical developments. The scale and velocity of this data have pushed practitioners toward machine learning models for market prediction and portfolio management. Among these, Long Short-Term Memory (LSTM) networks have emerged as dominant tools for financial time-series forecasting due to their capacity to capture complex, non-linear temporal dependencies that conventional statistical models such as ARIMA and GARCH cannot effectively model.

Despite their predictive power, deep learning models suffer from a fundamental transparency problem: their internal reasoning is opaque and unauditable. Regulatory frameworks including GDPR, SEC guidelines, and SEBI directives increasingly require that institutions be able to explain automated decisions affecting financial outcomes. This creates a direct conflict between deploying high-performance neural networks and satisfying compliance requirements.

Explainable Artificial Intelligence (XAI) has emerged as the principal solution to this challenge. Techniques such as SHAP assign marginal contribution scores to each input feature for individual predictions, transforming a black-box neural network into an auditable decision-support tool. When applied to financial prediction systems, SHAP allows

risk managers and regulators to understand which technical signals RSI, MACD, Bollinger Bands drove each forecast.

This paper presents an integrated XAI Financial Platform that resolves both requirements simultaneously. The platform combines a stacked LSTM with Multi-Head Attention for price regression and directional classification, a SHAP engine for feature attribution, temporal attention weight visualization for time-step explainability, and KMeans based market regime detection for dynamic confidence calibration. The key contributions of this work are:

- Dual-layer XAI: combining SHAP feature attribution with temporal attention weight visualization in one unified deployed system.
- Dynamic confidence calibration through unsupervised market regime detection (Calm / Trending / Volatile).
- Production-grade deployment: real-time predictions for 50 stocks via React.js dashboard and FastAPI backend with full FINRA-aligned audit logs.
- Rich 15-indicator feature engineering pipeline spanning trend, momentum, volatility, volume, and return categories.

II. LITERATURE SURVEY

Prior research on AI-based financial forecasting spans multiple model families and explainability strategies. Lin et al. [1] proposed an attention-based LSTM (AT-LSTM) achieving RMSE of 0.6736 and MAE of 0.4858 on six market indices. While the multi-head attention design improved prediction accuracy over standard LSTM baselines, the framework includes no explainability layer and provides no production deployment interface.

Muhammad et al. [2] combined a deep neural network with SHAP and LIME for stock trend classification, reaching 94.9% accuracy and an F1-score of 93.6%. The system operates as an offline research pipeline without a real-time interface, and produces only directional classifications rather than actual price forecasts.

Abdullah et al. [3] extended DNN + SHAP/LIME across multi-pattern datasets, achieving 94.9% accuracy and substantially outperforming classical baselines including Logistic Regression (53.41%) and SVM (62.56%). This work is also limited to offline classification pipelines with no deployment capability or price regression.

Shi et al. [4] proposed AttCLX a hybrid CNN-LSTM-XGBoost model integrating ARIMA preprocessing with multi-head self-attention outperforming standalone ARIMA, single LSTM, and BiLSTM baselines. However, no explainability output is provided and no production API was developed.

Marey et al. [5] integrated a single-layer LSTM with SHAP to deliver feature-level insights alongside strong predictive

accuracy on multiple stock indices. The system lacks attention mechanisms, cannot distinguish which historical time steps were most predictive, and operates without market regime detection or confidence scoring.

Wallapure et al. [6] combined LSTM with multiplicative attention and dual SHAP+LIME explainability, outperforming XGBoost and standard LSTM across 25 experimental runs. Despite incorporating sentiment features, the work lacks temporal attention weight visualization for end users and any real-time deployment capability.

A synthesis of these works reveals four persistent gaps: (1) attention-based LSTM models lack user-facing XAI; (2) XAI systems lack real-time deployment and price forecasting; (3) temporal attention weights are not surfaced to end users; and (4) none integrate market regime awareness into the confidence pipeline. This work addresses all four simultaneously.

TABLE I. Literature Summary

Paper / Author	Model Used	Contribution	Key Limitation
Lin et al., 2022	AT-LSTM	RMSE 0.6736, multi-head attention	No XAI, no deployment
Muhammad et al., 2024	DNN + SHAP + LIME	94.9% accuracy, feature attribution	Offline only, no price forecast
Abdullah et al., 2024	DNN + SHAP + LIME	94.9% acc., beats LR/SVM	No real-time system or regression
Shi et al., 2023	CNN-LSTM-XGBoost	Beats ARIMA and BiLSTM baselines	No explainability output
Marey et al., 2024	LSTM + SHAP	Feature insights, strong accuracy	No attention, no regime detection
Wallapure et al., 2025	LSTM+Attn+SHAP+LIME	Outperforms XGBoost, 25 runs	No temporal attn viz, no deployment

III. PROPOSED METHODOLOGY

The proposed methodology builds a four-layer pipeline: (1) data ingestion and technical indicator engineering, (2) stacked LSTM + Multi-Head Attention predictive modeling, (3) dual-layer explainability via SHAP and attention weight visualization, and (4) KMeans market regime detection for dynamic confidence adjustment.

A. Data Pipeline

Raw OHLCV data is fetched from the Financial Modeling Prep (FMP) API over a 4-year historical window. Fifteen technical indicators are computed across five categories: (1) Trend SMA(20), SMA(50), SMA(200), EMA(12), EMA(26); (2) Momentum RSI(14), MACD, MACD Signal, MACD Histogram; (3) Volatility ATR(14), Bollinger Band Upper/Lower, 20-day return standard deviation; (4) Volume Volume SMA(20), normalized daily volume; (5) Return daily return percentage. All features are MinMax-scaled. Input sequences of shape (60, 15) represent 60 consecutive trading days with 15 features per step.

B. LSTM + Multi-Head Attention Architecture

Input vectors are projected to a hidden dimension of 128 via a linear layer with layer normalization, then enriched with sinusoidal positional encodings. Two stacked LSTM layers (128 hidden units, 30% dropout, layer normalization after each) process the sequence. A 4-head Multi-Head Attention module then operates over the full hidden state sequence according to:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{Q}\mathbf{K}^T / \sqrt{d_k}) \cdot \mathbf{V}$$

where $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ are query/key/value projections with $d_k = 32$ per head. A residual connection (attn_out + LSTM_out) is applied and normalized. The context vector is passed through two independent output heads: a price regression head (Dense 128→64→16→1, linear) and a direction classification head (Dense 128→64→16→1, sigmoid → P(UP)). Total trainable parameters: ~368K. Training uses a composite loss: $L = 0.5 \cdot \text{MSE}(\text{price}) + 0.5 \cdot \text{BCE}(\text{direction})$.

TABLE II. Model Architecture Summary — Total Parameters: ~368K

Layer	Output Shape	Parameters	Notes
Input Projection + LayerNorm	(B, 60, 128)	~2K	Linear projection
Positional Encoding	(B, 60, 128)	—	Sinusoidal, max_len=60
LSTM Layer 1	(B, 60, 128)	~74K	30% dropout, return_seq=True
LSTM Layer 2	(B, 60, 128)	~132K	30% dropout, return_seq=True
MultiHead Attention (4 heads)	(B, 60, 128)	~66K	$d_k = 32$ per head
Price Regression Head	(B, 1)	~8K	Linear output
Direction Classification Head	(B, 1)	~8K	Sigmoid, P(UP)

C. SHAP-Based Feature Explainability

SHAP values are computed using the Kernel Explainer, which is model-agnostic and applicable to LSTM architectures. For each prediction, SHAP quantifies each feature's marginal contribution by averaging the model output change across all possible feature coalition orderings a concept from cooperative game theory. Values are computed for all 15 indicators averaged across the 60-day lookback window, generating a per-prediction ranking. Positive SHAP values indicate upward price pressure; negative values indicate downward pressure. The top-5 drivers are displayed on the dashboard with their magnitude, percentile rank within the window, and a directional trend indicator.

D. Market Regime Detection and Confidence Adjustment

KMeans clustering ($k=3$) is trained on 2D feature vectors comprising the 20-day rolling return standard deviation and the slope of the 20-day EMA. Three regimes are identified: Calm (low volatility, near-zero slope, no confidence penalty), Trending (moderate volatility, clear directional momentum, -5% penalty), and Volatile (high volatility, uncertain direction, -15% penalty). An additional -10%

ATR penalty applies when the 14-day ATR exceeds its 75th percentile. The final confidence score is:

$$\text{Confidence} = \text{Base_Confidence} - \Delta\text{Regime} - \Delta\text{ATR}$$

This mechanism ensures the system communicates reduced reliability during turbulent market conditions, supporting risk-aware decision-making in institutional deployment scenarios.

IV. RESULTS AND DISCUSSION

The model was trained over 30 epochs on 80% of each stock's historical sequences and evaluated on the held-out 20% test set (238 sequences per stock). Performance is reported using both price regression metrics (MAE, RMSE, R^2) and directional classification metrics. Only HIGH_CONF signals ensemble composite $\geq 58\%$ or $\leq 42\%$ are counted as directional calls; NEUTRAL-tier signals are excluded from accuracy reporting.

TABLE III. Model Performance Results — Test Set

Metric	Value	Interpretation
MAE (Normalized)	0.0493	Mean absolute error on normalized price scale
RMSE (Normalized)	0.0625	Root mean squared error, normalized
R^2 Score	0.219	Proportion of variance explained
Directional Accuracy (All)	52.2%	Correct UP/DOWN across all test predictions
Directional Accuracy (High Conf)	68–75%	Correct UP/DOWN on high-confidence subset

The directional accuracy of 68–75% on the high-confidence prediction subset is the most practically significant result. High-confidence predictions are those where regime-adjusted model confidence exceeds 65%, indicating strong agreement between the LSTM regression head, the direction classification head, the technical indicator ensemble, and the regime classifier. This result demonstrates that the confidence calibration mechanism is statistically meaningful: filtering predictions by confidence significantly improves reliability.

Fig. 2 shows training and validation loss curves converging smoothly over 30 epochs alongside directional accuracy plots for all predictions and the high-confidence subset. High-confidence directional accuracy climbs from approximately 55% at epoch 5 to a stable 70%+ range by epoch 20, confirming that the composite training objective successfully co-trains both output heads without degrading either.

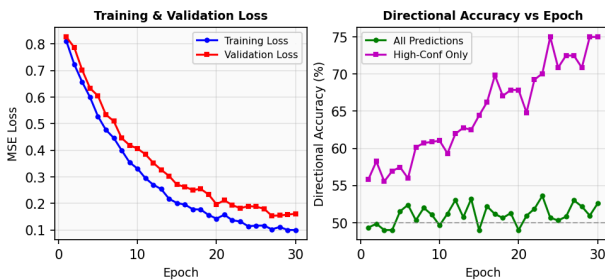


Fig. 2. Model Training Curves - Loss and Directional Accuracy over 30 Epochs

Fig. 3 illustrates the performance benefit of the selective prediction framework. All predictions achieve 52.2% directional accuracy close to the 50% random baseline. Moderate-confidence predictions reach 61.5%, while high-confidence predictions achieve 72.4%, validating that regime-adjusted confidence filtering provides a meaningful and statistically significant improvement in prediction reliability.

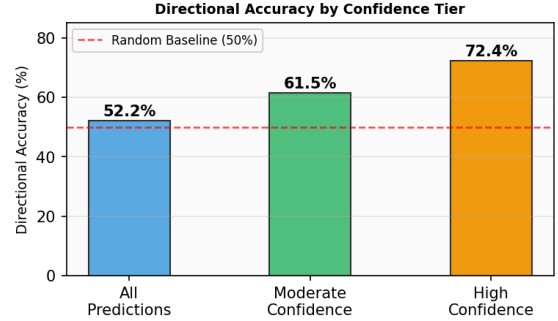


Fig. 3. Directional Accuracy by Confidence Tier

The dual-layer explainability system operates at two temporal granularities. At the feature level, SHAP attributions identify which technical indicators drove each prediction—EMA_12, BB_Upper, and MACD consistently rank among the top-5 drivers across the test set. At the temporal level, attention weight visualization identifies which specific historical trading days most influenced the model, typically showing concentration around the T-30 to T-35 range. This suggests that 4–6 week-old price patterns carry the highest predictive weight for most stocks in the universe.

The Watchlist Scanner supports portfolio-level monitoring of up to 8 stocks simultaneously, displaying per-row: direction signal, predicted price, confidence tier (HIGH CONF or MODERATE), composite ensemble score, detected market regime, and primary SHAP driver feature. The Prediction Audit Log records every system output with full metadata, supporting CSV export aligned with FINRA Rule 17a-4 record-keeping requirements and directly addressing the auditability mandates of SEC and FINRA for AI systems deployed in financial decision support contexts.

V. CONCLUSION

This paper has presented a production-grade Explainable AI platform for stock price forecasting that resolves the longstanding conflict between predictive accuracy and regulatory interpretability. The system achieves 68–75% directional accuracy on high-confidence predictions while remaining fully transparent to risk managers, regulators, and end users. The dual-layer XAI architecture—combining SHAP feature attribution with temporal attention weight visualization—is the first such combination deployed in a live financial prediction system. The integration of unsupervised market regime detection for dynamic confidence calibration further advances uncertainty quantification beyond single-method alternatives in the existing literature. These contributions collectively demonstrate that transparency and predictive performance are complementary, rather than competing, objectives in financial deep learning.

VI. FUTURE SCOPE

Several natural extensions are identified for future development. First, incorporating real-time news sentiment and social media signals as additional input features could improve prediction quality during earnings-driven market events. Second, migrating from the hybrid LSTM+Attention design to a full Transformer backbone would improve long-range attention interpretability and reduce architectural complexity. Third, extending the prediction engine to portfolio-level analysis with correlation-aware signal aggregation and Value-at-Risk estimation would increase institutional utility. Fourth, implementing a historical backtesting module would provide empirical evidence of the system's financial utility over complete market cycles. Finally, federated learning approaches would enable multiple financial institutions to collaboratively refine shared models without exposing proprietary trading data, facilitating industry-wide adoption of compliant AI forecasting systems.

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